

CLEANLOOP COMMUNITY

Smart Sustainable Public Toilets for Cleaner, Healthier
Communities

“Monitor. Maintain. Reuse. Sustain.”

Smart sensors

Water recycling

Rainwater harvesting

Map navigation



Rain · Sensors · Map

The Problem: Public Toilets Today

Key problems

- Inconsistent cleaning and maintenance
- Lack of soap, toilet paper and water
- Delayed detection of odour and hygiene issues
- Water wastage in handwashing and cleaning
- Poor management of wastewater
- No real-time toilet availability information
- Unpleasant surroundings, little greenery

Today: neglected

- Empty dispensers, dirty floors
- Overflowing drains, bad odour

With CLEANLOOP: clean & green

- Monitored, verified, restocked
- Recycled water, plants, shade

Our Solution: CLEANLOOP

CLEANLOOP adds a digital monitoring and water-management layer to existing public toilet infrastructure.

1 Smart toilet monitoring

2 Consumable tracking

3 Odour / air-quality indication

4 QR-based cleaning verification

5 Citizen map navigation

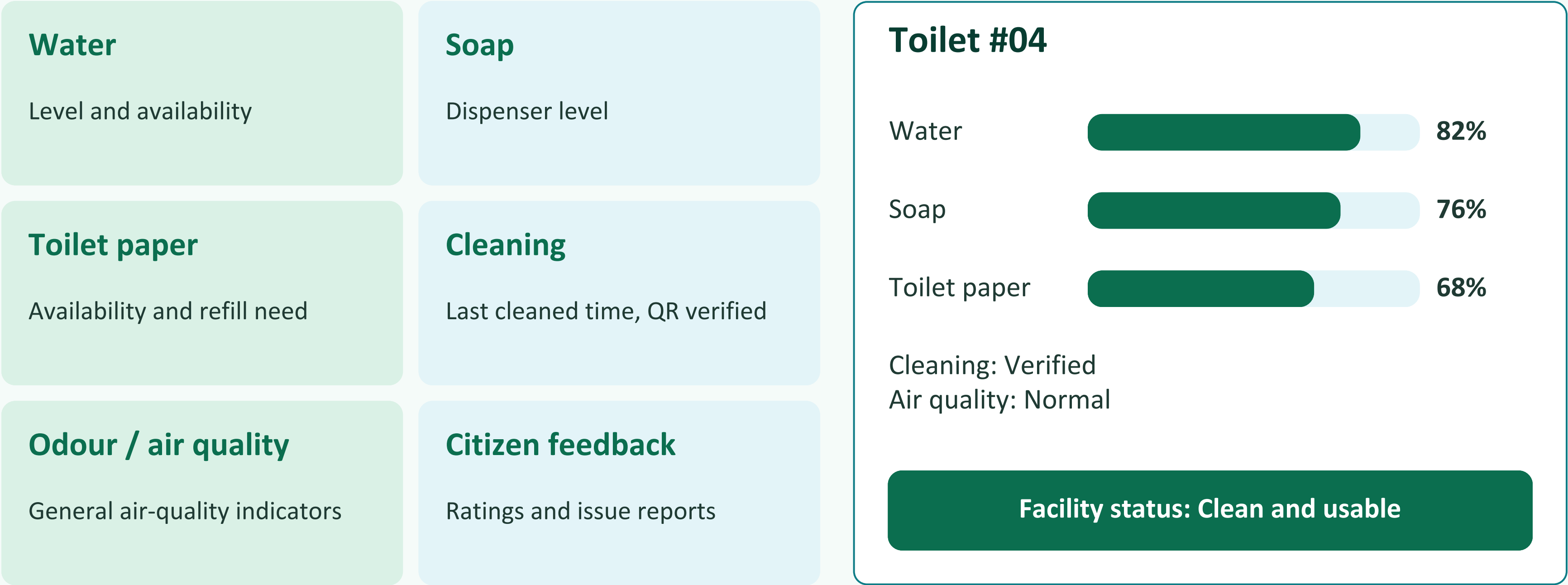
6 Greywater treatment and reuse

7 Rainwater harvesting

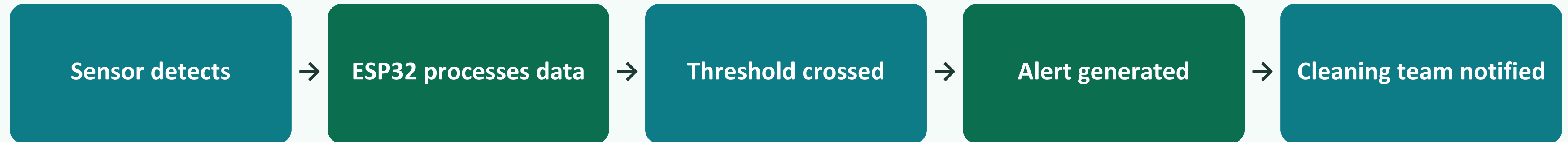
8 Green landscaping

9 Sewer and wastewater management

Smart Toilet Monitoring System



Odour and Air-Quality Monitoring



Low-cost components

- MQ-135 or similar sensor (prototype)
- ESP32 microcontroller
- Data sent to the dashboard
- Predefined prototype thresholds
- Alert for unusual readings

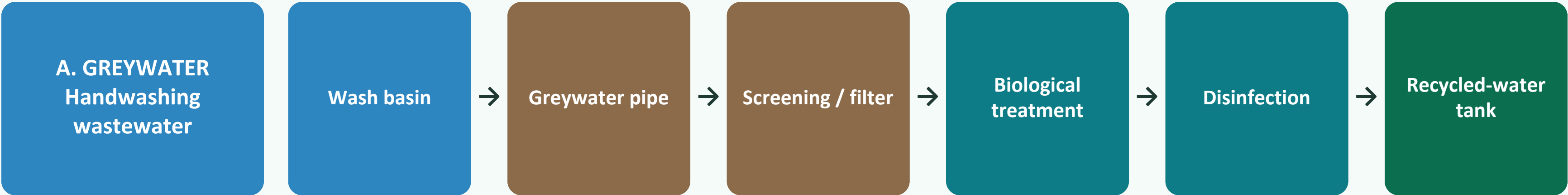


Toilet #04 — Air-quality/odour alert

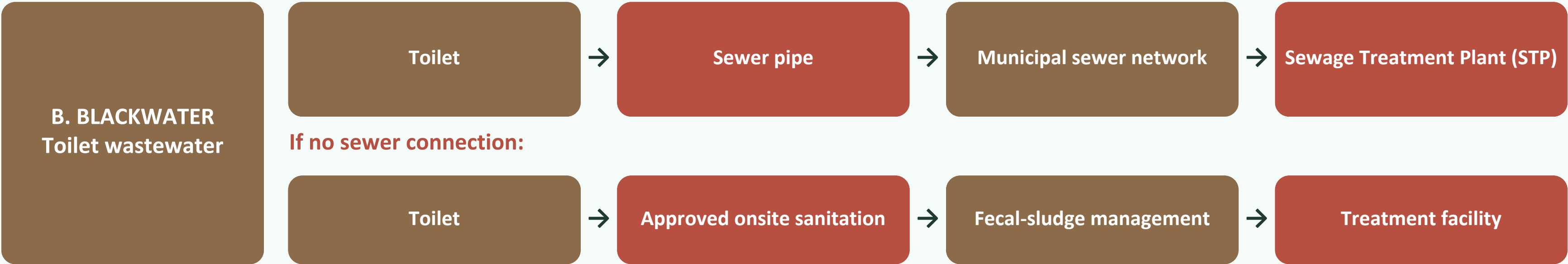
Cleaning request sent

The prototype indicates air-quality changes and possible odour concerns. It does not independently decide whether a toilet is clean.

Sustainable Water Management

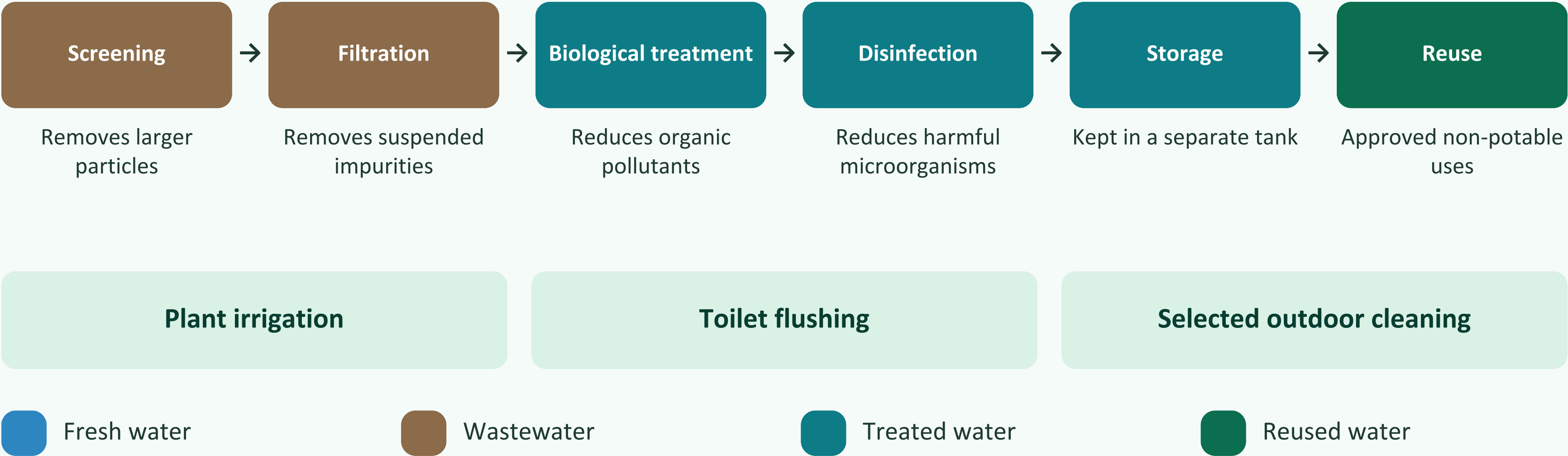


Non-potable uses: toilet flushing · irrigation of suitable plants · outdoor cleaning where appropriate



Greywater and blackwater are always handled separately.

Water Purification and Reuse



Safety: recycled greywater must not be used directly for handwashing or drinking unless treated and quality-monitored for that purpose.

Green and Sustainable Surroundings

The toilet becomes a clean, green and welcoming community facility.

Treated water is used only where suitable for the plants and irrigation system.

Native, low-maintenance plants

Shaded seating area

Rainwater harvesting

Solar panels for sensors and lighting

Permeable paving

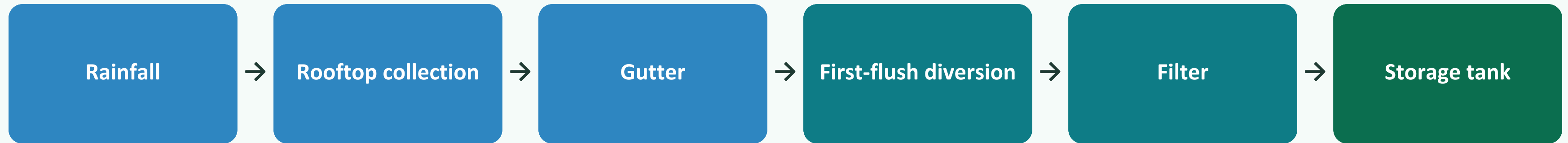
Proper drainage

Accessible pathways

Natural ventilation and lighting

Clean walking space around the toilet

Rainwater Harvesting



Water-saving benefits

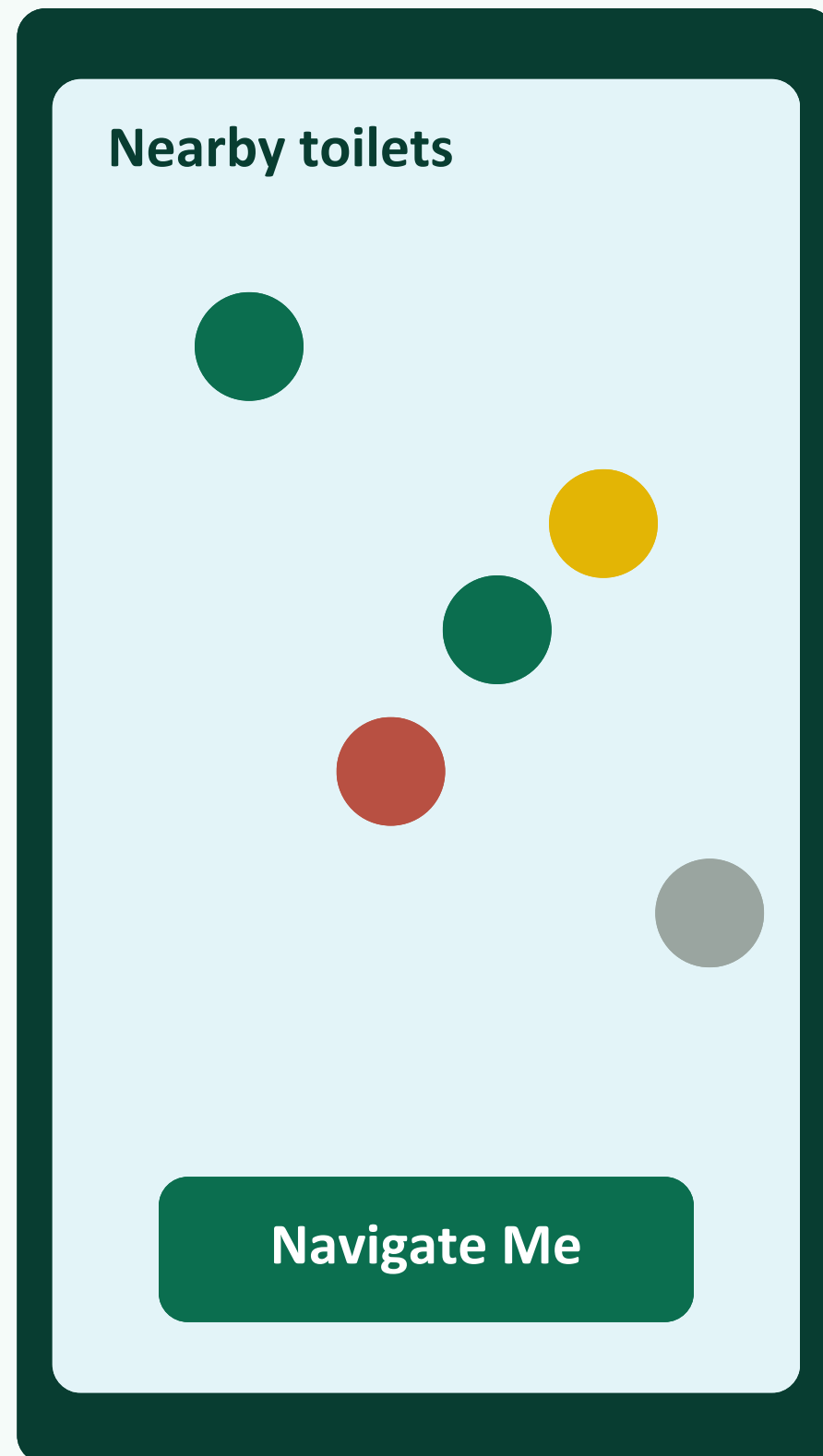
- Less dependence on fresh water
- Less stormwater runoff
- Supports plant growth
- Better local water management
- Helps conserve water resources

Potential uses

- Toilet flushing · irrigation
- Outdoor cleaning (per local rules)

Harvested rainwater is not automatically safe for drinking or handwashing.

Citizen Map Navigation and Smart Alerts



Citizen features

- View nearby public toilets
- Find nearest usable toilet
- Tap “Navigate Me”
- See water, soap, cleaning status
- Scan QR code
- Submit feedback or report problems

Worker features

- Receive cleaning alerts
- View assigned tasks
- Scan QR after cleaning
- Upload before/after photos
- Update facility status

-  Green: Clean and usable
-  Red: Maintenance required

-  Yellow: Attention required
-  Grey: Temporarily unavailable

Complete CLEANLOOP Workflow



- Sensors and QR codes collect information
- Alerts are sent when attention is needed
- Cleaning staff receive assigned tasks
- Staff verify the completed work
- The public map is updated
- Greywater and rainwater are reused for suitable purposes
- Data reveals recurring maintenance issues

Innovation, Impact and Future Scope

Key innovation

- Consumable monitoring
- Citizen navigation
- Cleaning verification
- Water conservation
- Greywater reuse
- Green surroundings
- Digital maintenance tracking

Expected impact

- Better user experience
- Reduced water wastage
- Faster maintenance response
- Improved transparency
- More sustainable infrastructure
- Cleaner, greener communities

Future scope

- AI maintenance prediction
- Solar-powered monitoring
- Better water-quality sensors
- Municipal dashboards
- Accurate sensor calibration
- Schools, hospitals, bus stands, parks

CLEANLOOP transforms a public toilet from a basic facility into a monitored, sustainable and community-friendly infrastructure system.

Monitor. Maintain. Reuse. Sustain.

Smart Toilet Monitoring System

Real-time monitoring of toilet consumables and cleanliness for better hygiene, improved user experience and efficient maintenance — all within the CleanLoop ecosystem.

Focus on Toilet Essentials



Water



Soap



Toilet Paper



Cleaning
Status

How It Works



Water Sensor

- Monitors water level in tank & usage
- Detects leaks / abnormal flow



Soap Level Sensor

- Checks soap dispenser level
- Alerts for refill



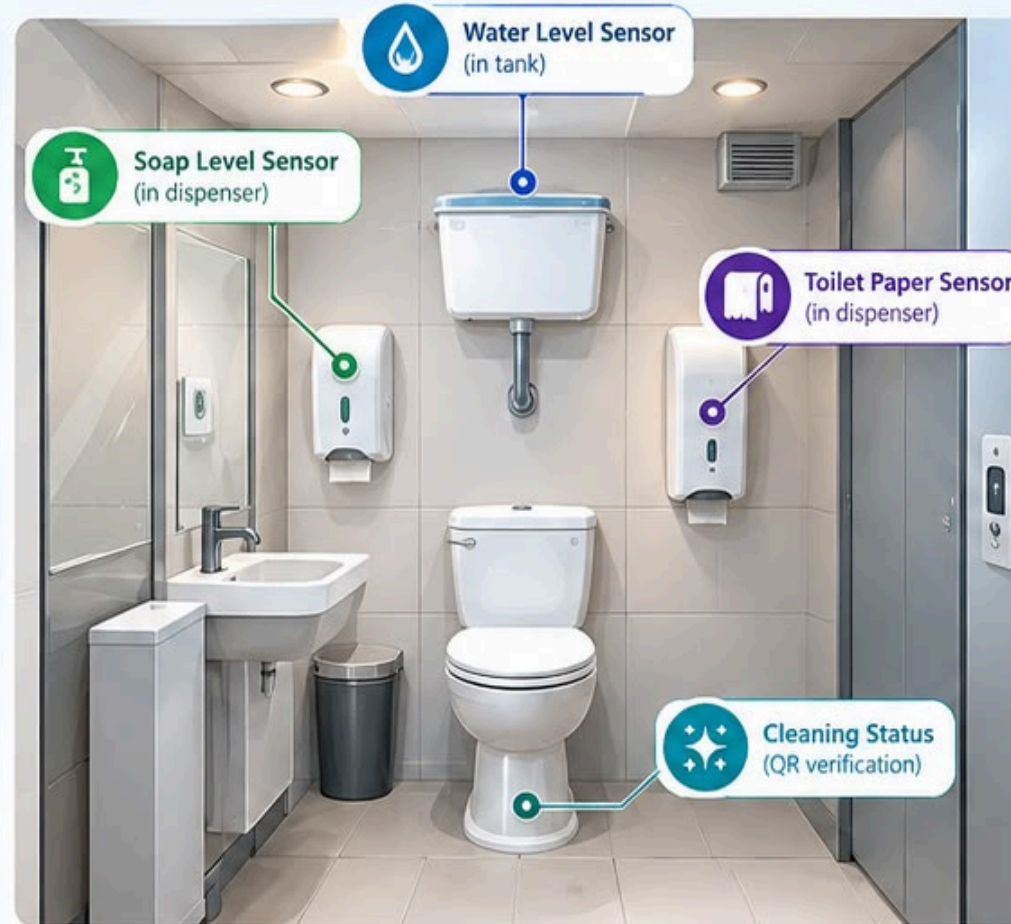
Toilet Paper Sensor

- Monitors roll level
- Notifies when to refill



Cleaning Status

- Tracks last cleaning time
- Links with QR verification from cleaning staff



Live Status – Toilet #04

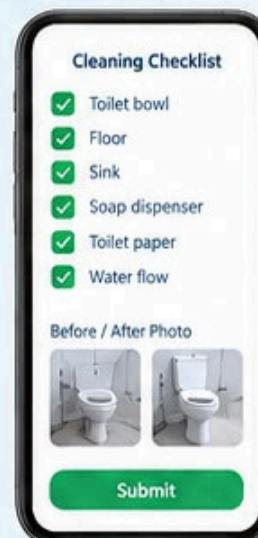
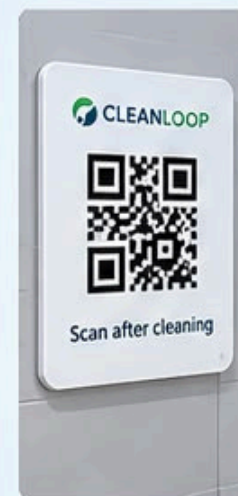
Toilet Health Score

92/100 food

Water Level	Water	82%
Soap Level	Soap	76%
Toilet Paper	Paper	68%
Last Cleaning	10:30 AM	Clean
Air Quality (Odour)	Normal	Normal
Citizen Feedback	4.5/5	★★★★★

✓ Facility Status: Clean & Usable

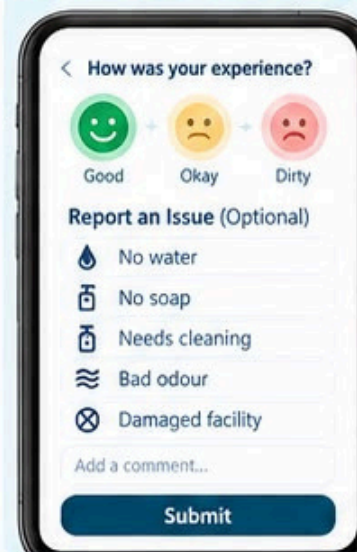
QR Code for Cleaning Staff



Worker scans the QR code after cleaning and updates the status with photo proof.

✓ Verified & Updated on Map

Citizen Feedback & Reporting



Scan the QR code at the toilet to give feedback or report any issue.

Location is automatically attached to your report.

Smart Alerts & Notifications

Water Level Low	Toilet #07 – 22%	Refill Required
Soap Low	Toilet #04 – 15%	Refill Soon
Toilet Paper Low	Toilet #12 – 10%	Refill Required
Cleaning Overdue	Toilet #09 – 2 hrs	Cleaning Needed
Air Quality Normal	Toilet #06	All Good

The CleanLoop Toilet Loop

A smarter way to keep public toilets clean, monitored and ready.



Monitor
(Sensors + QR)



Detect
(Alerts & Data)



Assign
(Worker Task)



Clean
(Verification)



Update
(Map Status)



Healthy
(Community Use)

Cleaner Toilets
Healthier Communities
A Sustainable Tomorrow

1 Employment Working Process



Employment Structure

(For 10 Public Toilets – Illustrative)

	Sanitation Workers Daily cleaning, restocking, hygiene checks and shift coverage.	10 – 15
	Supervisors Work allocation, safety checks, quality control and issue escalation.	1 – 2
	Maintenance Technicians Plumbing, electrical, sensor and water-system maintenance. Can be shared across multiple toilets.	1 – 2
	Digital Support Dashboard monitoring, reports, scheduling and support. Can be shared across locations.	1

Total Illustrative Workforce 14 – 21 people

These are planning estimates, not a confirmed staffing requirement. Adjust based on operating hours, workload and applicable labor requirements.

Career Growth Path

- Sanitation Worker**
Starts with hygiene training and daily cleaning responsibilities.
- Senior Sanitation Worker**
Gains experience in quality checks, supply tracking and assisting new staff.
- Toilet Supervisor**
Takes responsibility for team scheduling, inspections and service reporting.
- Operations Coordinator**
Manages multiple locations, staffing, dashboards and client communication.

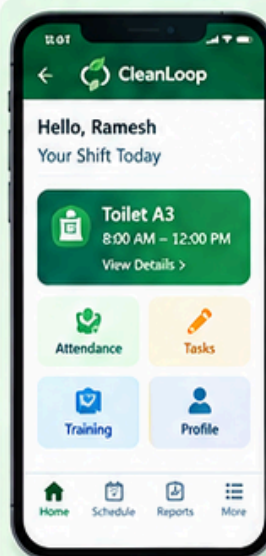
Additional Employment Opportunities

- ✓ Plumbing & electrical service providers
- ✓ Water recycling system maintenance
- ✓ Sensor installation & technical support
- ✓ Training and safety coordinators
- ✓ Data entry & cleanliness report operators

Career progression depends on available positions, skills, training and fair promotion practices — not just time served.

CleanLoop Worker Connect

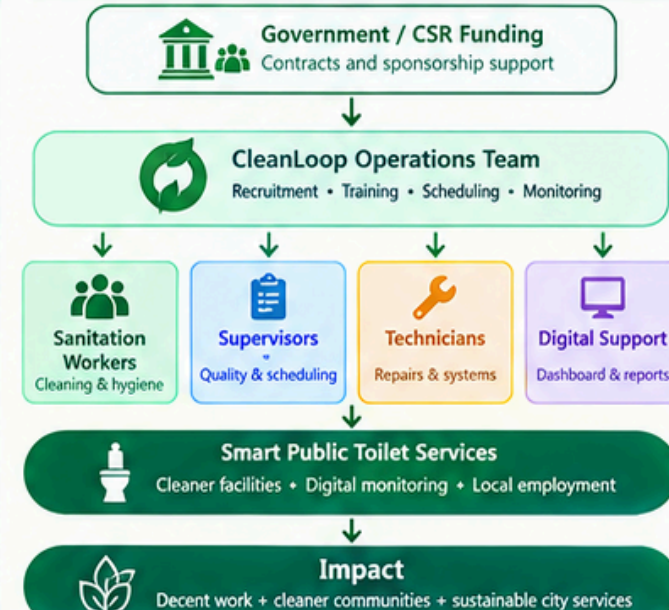
A digital employment and workforce management system



How It Works

- Local Worker Registration**
Workers can register through the platform or an authorized recruitment partner.
- Skill Training**
Hygiene, safety, cleaning equipment and digital tools training.
- Smart Work Allocation**
Workers receive shifts and cleaning tasks based on location and availability.
- Transparent Payroll**
Attendance records and approved hours support accurate salary processing.
- Skill & Career Growth**
Track completed training and support promotion into senior roles where opportunities exist.

The Employment Ecosystem



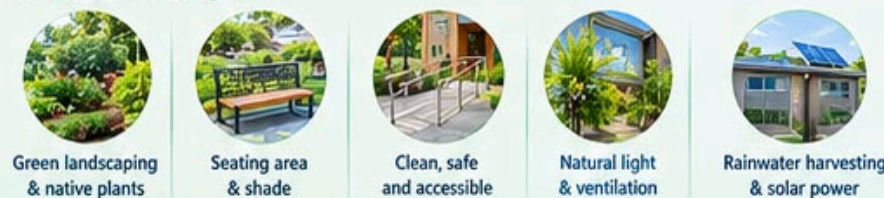
*A cleaner toilet
is not just a facility,
it's a healthier community
and a greener planet.*



Inside the System – How It Works



The Surroundings



Water Resource Utilization



How It All Connects



How Water is Managed & Recycled

A closed-loop system for smarter water use.

1 Hand Wash Water (Greywater)



2 Toilet Wastewater (Blackwater)



OR (if no sewer connection)



3 Water Reuse in the Surroundings



Existing Projects & Inspiration

Real initiatives are already improving public sanitation in India and abroad

Insert official photo

EXISTING INITIATIVE

NDMC Smart Public Amenities Centers

 **New Delhi, India**

Toilet, water ATM, cash machine and digital clinic in one; solar panels and feedback tablets

Insert official photo

EXISTING INITIATIVE

LooCafe

 **Hyderabad, India**

Toilets with a business model, IoT monitoring and water recycling; also in other Indian cities

Insert official photo

EXISTING INITIATIVE

Pattana Pragathi Toilet Monitoring System (PPTMS)

 **Telangana, India**

Mobile inspections and a real-time dashboard for public and community toilets

Insert official photo

EXISTING INITIATIVE

Singapore Smart Toilet Monitoring

 **Singapore**

People counters, ammonia, humidity and wetness sensors and supply monitors guide cleaning

Insert official photo

EXISTING INITIATIVE

Ti (Toilet Integration) by Saraplast

 **Pune, India**

Women's restrooms in refurbished buses, with digital feedback panels and water-efficient fixtures



CleanLoop Community builds on these by combining air-quality indication, consumable tracking, QR cleaning verification, citizen navigation, greywater reuse, rainwater harvesting and local employment in one low-cost model.